

FashionRec

Amazon Fashion Recommendation Explorer

A from-scratch collaborative, content-based, and hybrid recommendation system with explainability, a full evaluation suite, and a browser-based UI.

Course: Recommender Systems

Dataset: Amazon Fashion 5-Core (JSONL)

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Stack: Python 3 · Flask · Vanilla JS

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1 Problem & Motivation

Online fashion retail generates enormous product catalogues. A typical shopper encounters hundreds of items without any personalised guidance, making discovery slow and frustrating. The core problem this project addresses is: given a user's historical ratings of Amazon fashion products, how can the system rank unrated items so that the most relevant ones appear at the top — and how can it explain why each item was selected?

Recommendation systems tackle this problem through two broad families of signals: (1) collaborative signals, derived from the behaviour of similar users or items, and (2) content signals, derived from item attribute metadata. A pure collaborative approach suffers from the cold-start problem; a pure content approach ignores community taste patterns. This project implements and systematically compares both families, then blends them in a hybrid model.

Beyond accuracy, the project demands transparency: every recommendation exposes the evidence behind it — which users, items, or attributes drove the score — so that the system is auditable and trustworthy. Explainability is baked into every model, not added as an afterthought.

2 Dataset & Data Sources

The project uses the Amazon Fashion 5-Core dataset distributed as a JSONL file (AMAZON_FASHION_5.jsonl). The '5-Core' designation means that every user and every item in the subset has at least five interactions, ensuring a minimum density for collaborative methods.

2.1 Dataset Statistics

Statistic	Value
Total review records	3,176
Unique users (reviewerID)	406
Unique items (ASIN)	31
Avg reviews per user	7.82
Avg reviews per item	102.45
Max reviews by one user	16
Max reviews on one item	377
Matrix sparsity	74.77 %
Verified purchase %	96.97 %
Records with style data	3,107 (97.8 %)
Overall avg rating	4.404 / 5.0

2.2 Rating Distribution

The dataset exhibits a strong positive skew — a common property of e-commerce review data. The vast majority of ratings are 5-star, reflecting purchase-confirmation bias.

Rating	Count	Percentage
--------	-------	------------

★☆☆☆ (1)	117	3.68 %
★★☆☆ (2)	93	2.93 %
★★★☆☆ (3)	337	10.61 %
★★★★☆ (4)	471	14.83 %
★★★★★ (5)	2,158	67.95 %

2.3 Record Schema

Each JSONL line is a JSON object with the following fields:

reviewerID — String — unique user identifier (e.g. 'ALJ66O1Y6SLHA')

asin — String — Amazon Standard Identification Number (item ID)

overall — Float — star rating on the 1–5 scale

reviewerName — String — human-readable display name

verified — Bool — whether the purchase was verified

style — Dict — key/value pairs for Size, Color, Size Name, Style

reviewText — String — free-text review body (not used in models)

summary — String — review headline

unixReviewTime — Int — POSIX timestamp of the review

The style attribute is the backbone of the content-based approach. Four distinct style keys appear in the dataset: Color, Size, Size Name, and Style. Each (key, value) pair is converted into a binary feature flag during matrix construction (e.g. 'Color:Blue/Orange' = 1.0).

3 System Architecture

FashionRec is structured as a three-layer system: a data layer, a recommendation engine layer, and a presentation layer. The figure below describes the end-to-end pipeline.

3.1 End-to-End Pipeline

[1] **JSONL File** — AMAZON_FASHION_5.jsonl — raw review records stored on disk.

[2] **load_data()** — Reads every line, parses JSON, and returns a list of raw record dicts.

[3] **build_matrices()** — Constructs five in-memory structures: user_item (user→item→rating), item_user (item→user→rating), item_meta (item→feature→1.0), user_means, and item_means.

[4] **Model Layer** — Four recommender classes — UserBasedCF, ItemBasedCF, ContentBasedFilter, HybridRecommender — operate on these matrices.

[5] **Flask API (app.py)** — Exposes three REST endpoints: /api/users, /api/user/<id>/history, and /api/recommend. Models are instantiated per request with the caller-specified similarity and prediction parameters.

[6] **Browser UI (index.html)** — Single-page application. Fetches /api/users on load, renders a scrollable user list, and calls /api/recommend when the user clicks 'Get Recommendations →'.

3.2 File Structure

File	Role
recommender.py	Core recommendation engine — all models, similarity functions, prediction strategies, evaluation framework, and Explanation class
app.py	Flask web server — REST API routes, data loading at boot, serialisation of Explanation objects to JSON
get_recommendations.py	Standalone CLI script — loads data and runs UserCF with Pearson/mean-centred for a hardcoded user ID
static/index.html	Self-contained SPA — all HTML, CSS, and JavaScript in a single file; no build step required
AMAZON_FASHION_5.jsonl	Dataset — 3,176 Amazon Fashion reviews in JSON Lines format

4 Core Components

4.1 User-Based Collaborative Filtering (UserBasedCF)

Class `UserBasedCF` in `recommender.py` implements memory-based user-user collaborative filtering. The key assumption is that users who agreed in the past will agree in the future.

Algorithm overview:

- For a target user u , compute the similarity between u and every other user in the dataset.
- Select the top- K most similar users (neighbours) where K defaults to 20.
- For each candidate item (items u has not yet rated), aggregate the neighbours' ratings using the configured prediction function (weighted average, mean-centred, or significance weighting).
- Sort candidates by predicted score and return the top- N .

Key parameters exposed at construction time:

Parameter	Default	Options
similarity	cosine	cosine · pearson · euclidean
prediction	mean_centered	weighted_average · mean_centered · significance_weighting
top_k	20	any positive integer

Similarity scores are cached in a dict keyed by the sorted (u_1, u_2) tuple to avoid redundant computation across repeated calls.

4.2 Item-Based Collaborative Filtering (ItemBasedCF)

Class `ItemBasedCF` mirrors the structure of `UserBasedCF` but pivots the similarity computation to the item space. Two items are similar if the users who rated both gave consistent ratings. The advantage over user-based CF is greater stability: item relationships change more slowly than user behaviour.

For a target user u and candidate item i , the model finds the K items most similar to i , filters down to those the user has already rated, and applies the prediction formula over that rated-neighbour set. The same three similarity and three prediction strategies are supported.

4.3 Content-Based Filtering (ContentBasedFilter)

Class `ContentBasedFilter` operates entirely on item metadata, bypassing the rating matrix. It does not compute user-to-user or item-to-item similarities; instead it models user taste as a weighted average of the feature vectors of items the user has rated.

Profile construction formula:

```
profile[feat] += rating × item_meta[item][feat]
```

```
profile = { feat: val / total_weight for each feat }
```

At recommendation time, each unseen item's feature vector (binary flags for its style attributes) is compared to the profile using cosine similarity. Items whose attributes most closely match the user's historical preferences rank highest.

Because it relies only on metadata, the content-based filter can generate recommendations for items with zero rating history — partially addressing the item cold-start problem.

4.4 Hybrid Recommender (HybridRecommender)

Class HybridRecommender combines the three models above using a weighted linear blend of their normalised scores. Each sub-model independently produces a ranked list; scores are min-max normalised to [0, 1] before blending, ensuring no single model dominates due to scale differences.

Blend formula:

```
score(i) = w_user × UCF_norm(i) + w_item × ICF_norm(i) + w_content ×  
CBF_norm(i)
```

Default weights used in production and evaluation:

Component	Weight
UserBasedCF (UCF)	0.40 (40 %)
ItemBasedCF (ICF)	0.40 (40 %)
ContentBasedFilter (CB)	0.20 (20 %)

The pool size for each sub-model is $\max(n \times 5, 50)$ to ensure sufficient candidate overlap before blending. Explanation objects generated by the hybrid model identify the dominant contributing sub-model and include its sub-explanation automatically.

5 Similarity & Prediction Functions

5.1 Similarity Measures

All three similarity functions operate on sparse rating vectors represented as Python dicts. They are registered in the SIMILARITY_FUNCS dict and selected at model construction time.

Cosine Similarity

$$\text{cos}(a, b) = \sum (a[k] \times b[k]) / (\|a\| \times \|b\|)$$

Measures the angle between two vectors in item/user space. Does not account for rating scale bias (e.g. a user who always rates 5 and one who always rates 3 will appear highly similar).

Pearson Correlation Coefficient

$$\text{pearson}(a, b) = \sum ((a[k] - \bar{a})(b[k] - \bar{b})) / (\sigma_a \times \sigma_b)$$

Computed only on co-rated items (minimum 2 required). Removes individual rating-scale bias by mean-centering before computing the correlation. Generally preferred for rating-prediction tasks.

Euclidean Distance Similarity

$$\text{euclidean}(a, b) = 1 / (1 + \sqrt{\sum (a[k] - b[k])^2})$$

Transforms Euclidean distance into a similarity in (0, 1]. Simple and fast, but sensitive to magnitude differences and treats unrated items as zero, which can distort the distance in sparse settings.

5.2 Prediction Strategies

All three strategies aggregate neighbour ratings to produce a predicted score for a (user, item) pair. They are registered in PREDICTION_FUNCS.

Weighted Average

$$\text{pred} = \sum (\text{sim}(n) \times r(n)) / \sum |\text{sim}(n)|$$

The simplest approach — a similarity-weighted mean of neighbour ratings. Falls back to the global mean if no neighbour has rated the target item. Susceptible to rating-scale bias.

Mean-Centred (Bias-Adjusted)

$$\text{pred} = \mu_{\text{target}} + \sum (\text{sim}(n) \times (r(n) - \mu_n)) / \sum |\text{sim}(n)|$$

Corrects for per-user (or per-item) rating bias by computing deviations from each neighbour's mean before aggregating. Falls back to the target mean if no signal is available. This is the default and generally the best-performing strategy.

Significance Weighting

```
scale(n) = min(|common(target,n)|, T) / T    (T = min_common = 5)
adj_sim(n) = sim(n) * scale(n)
pred =  $\mu_{\text{target}} + \frac{\sum(\text{adj\_sim}(n) \times (r(n) - \mu_n))}{\sum \text{adj\_sim}(n)}$ 
```

Penalises neighbours who share very few co-rated items with the target, reducing overconfidence in similarity estimates derived from sparse overlap. Co-rated item counts are registered globally via `register_cocount()` at similarity computation time and stored in the module-level `_SW_cocount` dict.

6 Explainability Module

Every recommender class implements an `explain(user_id, item_id)` method that returns an Explanation object. This design makes explainability a first-class feature rather than an optional add-on.

6.1 Explanation Class

Attribute	Type	Description
item_id	str	ASIN of the recommended item
score	float	Predicted score / similarity used for ranking
method	str	One of: UserCF, ItemCF, ContentBased, Hybrid
reason	str	One-line plain-English summary shown to the user
details	dict	Method-specific supporting evidence (serialised to JSON by the API)

6.2 Evidence Surfaced by Each Model

UserCF: Top-3 most similar neighbours who rated this item; their similarity score, their rating of the item, and the number of items they share with the target user. Also reports the prediction method and clipped predicted score.

ItemCF: Top-3 most similar items the user has already rated; their similarity to the candidate, the user's own rating of each, and the number of shared raters between the candidate and each similar item.

ContentBased: Top-5 matching style features between the item vector and the user's taste profile (e.g. Color = 'Blue/Orange', profile weight = 0.412); top-3 past items that contributed most to building those feature preferences.

Hybrid: Normalised score contribution from each sub-model; the blended score; which sub-model was the primary driver; and the dominant sub-model's own explanation.

The UI renders explanations through a collapsible 'WHY THIS ITEM?' panel on each recommendation card, toggled client-side without an additional API call.

7 Evaluation Framework

7.1 Train / Test Split

The evaluation uses a per-user random hold-out split implemented in `train_test_split()`. For each user, 20 % of their rated items are withheld as the test set (minimum 1 item per user). A fixed random seed (`seed=42`) ensures reproducibility.

```
train_ui, test_pairs = train_test_split(user_item, test_ratio=0.2, seed=42)
```

After splitting, the training matrices are rebuilt from scratch using only the training records, so test-set items never leak into the similarity or mean computations.

7.2 Rating Prediction Metrics

`evaluate_rating_prediction()` iterates over all (user, item, true_rating) test triples. Predictions are clipped to [1, 5] before computing errors; failed predictions fall back to the midpoint (3.0).

RMSE $\sqrt{\sum(\text{pred} - \text{true})^2 / N}$ — Root Mean Squared Error — penalises large deviations more heavily. Lower is better.

MAE $\sum|\text{pred} - \text{true}| / N$ — Mean Absolute Error — the average magnitude of error. Lower is better.

7.3 Ranking Metrics

`evaluate_ranking()` samples up to `n_users=30` eligible users and generates top-K=10 recommendations for each. Held-out items serve as ground truth.

Precision@K Fraction of the top-K recommendations that appear in the held-out set.

Recall@K Fraction of the held-out items that appear in the top-K recommendations.

F1@K Harmonic mean of Precision@K and Recall@K.

Coverage Fraction of all training items that appear in at least one recommendation list.

7.4 Experimental Grid

The `run_experiments()` function evaluates all combinations of 3 similarity functions × 3 prediction strategies for both UserCF and ItemCF (18 configurations each = 36 total CF experiments), plus ContentBased and the Hybrid. For the Hybrid, the best-performing individual models are selected: Pearson/mean-centred for UserCF and Cosine/mean-centred for ItemCF.

8 Web Application & User Interface

8.1 Flask REST API

app.py starts a Flask application that loads the dataset once at boot and exposes three JSON endpoints:

Endpoint	Method	Description
GET /api/users	GET	Returns all users sorted by rating count; includes name, avg_rating, num_ratings
GET /api/user/<user_id>/history	GET	Returns a user's complete rating history sorted by rating (descending)
GET /api/recommend	GET	Accepts user_id, method, similarity, prediction, n; returns a list of Explanation dicts
GET /	GET	Serves static/index.html

Each recommendation object in the /api/recommend response is augmented with item-level metadata before serialisation: item_styles (the raw style dict), avg_rating (community average), and num_ratings (number of reviews), enabling the UI to display rich item cards without a second API call.

8.2 Single-Page Application

The frontend is a zero-dependency SPA contained in a single static/index.html file. It requires no build step, bundler, or CDN beyond the Google Fonts stylesheet.

Key UI features:

- Two-column layout: a fixed 320 px sidebar for user selection and settings, and a main panel for results.
- Live search across user names and IDs with instant re-render.
- Algorithm control panel: dropdown for approach (UserCF / ItemCF / Content / Hybrid), similarity measure, prediction function, and a range slider for result count (1–10).
- User profile strip showing initials avatar, display name, reviewer ID, total reviews, and average rating.
- Rating history row showing the last 8 rated items with star rating, ASIN, and style attributes.
- Recommendation grid with animated cards that each include: rank number, ASIN badge, method chip (colour-coded by model), style attribute tags, community rating, predicted score pill, and a score-progress bar.
- Collapsible 'WHY THIS ITEM?' explanation panel on each card — rendered client-side from the details dict without an additional API call.

- Toast notifications for network errors or invalid states.
- Responsive grid: cards re-flow to fill available width using CSS grid with `minmax(280px, 1fr)` columns.

Design system:

Font stack DM Serif Display (headings) · DM Sans (body) · DM Mono (code/metadata)

Colour palette Warm paper (#F5F0E8), dark ink (#0D0D0D), accent red (#C84B2F), blue (UserCF), green (ItemCF), gold (ContentBased)

Animation fadeUp keyframe (opacity 0→1, translateY 16px→0) staggered across cards; slideDown for explanation panels

9 Tech Stack Summary

Layer	Technology	Usage
Language	Python 3	Entire backend — models, API, CLI
Web Framework	Flask	REST API server; served via <code>app.run(debug=True, port=5000)</code>
Data Format	JSON Lines (JSONL)	Dataset storage — one review record per line
Numerics	math (stdlib)	sqrt, cosine, Pearson, Euclidean — no NumPy dependency
Utilities	collections (stdlib)	defaultdict for sparse matrix construction
Utilities	itertools (stdlib)	combinations — available but not used in current production path
Utilities	random (stdlib)	Seeded RNG for reproducible train/test splits and user sampling
Frontend	Vanilla JavaScript (ES2020)	SPA logic — fetch API, DOM manipulation, no frameworks
Styling	CSS3 (custom properties)	Design tokens via <code>:root</code> variables; CSS grid layout
Fonts	Google Fonts CDN	DM Serif Display, DM Sans, DM Mono
Serialisation	json (stdlib)	JSONL parsing; Flask's jsonify for API responses

Notably, the project is entirely stdlib + Flask for the backend — no NumPy, Pandas, Scikit-learn, or any ML framework is used. Every similarity function, prediction formula, and evaluation metric is implemented from mathematical first principles in pure Python.

10 Performance & Results

The following tables present the results produced by `run_experiments()` on the full 3,176-record dataset with an 80/20 train-test split (seed=42), K=10 for ranking metrics, and 30 sampled users for the ranking evaluation. RMSE and MAE are computed over all test pairs (~635 held-out ratings).

10.1 User-Based CF — All Configurations

Lower RMSE/MAE is better. Higher Precision@10, Recall@10, F1@10, Coverage is better.

Similarity	Prediction	RMSE	MAE	P@10	R@10	F1@10	Coverage
cosine	weighted_average	1.0877	0.8290	0.0667	0.6667	0.1212	0.6786
cosine	mean_centered	0.2096	0.0320	0.0633	0.6333	0.1152	0.6786
cosine	significance_weighting	0.4743	0.1305	0.0600	0.6000	0.1091	0.5357
pearson	weighted_average	1.0400	0.8131	0.0633	0.6333	0.1152	0.6429
pearson	mean_centered	0.2096	0.0320	0.0633	0.6333	0.1152	0.6429

pearson	significance_weighting	0.2096	0.0320	0.0633	0.6333	0.1152	0.6429
euclidean	weighted_average	0.5071	0.1189	0.0567	0.5667	0.1030	0.7857
euclidean	mean_centered	0.2096	0.0320	0.0633	0.6333	0.1152	0.6786
euclidean	significance_weighting	1.7108	1.5567	N/A	N/A	N/A	0.0000

Best UserCF configuration by RMSE: Cosine / Mean-Centred and Pearson / Mean-Centred both achieve RMSE = 0.2096, MAE = 0.032. Significance weighting with Euclidean similarity collapses (RMSE = 1.7108) because near-zero adjusted similarities cause division-by-zero fallback for most test pairs. N/A in P@10/R@10/F1@10 indicates no recommendations were generated for any sampled user under that configuration.

10.2 Item-Based CF — All Configurations

Similarity	Prediction	RMSE	MAE	P@10	R@10	F1@10	Coverage
cosine	weighted_average	0.2852	0.0526	0.0567	0.5667	0.1030	0.8214
cosine	mean_centered	0.3278	0.0730	0.0333	0.3333	0.0606	0.6071
cosine	significance_weighting	1.7297	1.5827	N/A	N/A	N/A	0.0000
pearson	weighted_average	0.3400	0.0738	0.0567	0.5667	0.1030	0.7500
pearson	mean_centered	0.3657	0.0920	0.0333	0.3333	0.0606	0.4643
pearson	significance_weighting	1.7276	1.5808	0.1000	1.0000	0.1818	0.3571
euclidean	weighted_average	0.2852	0.0526	0.0567	0.5667	0.1030	0.8214
euclidean	mean_centered	0.3278	0.0730	0.0333	0.3333	0.0606	0.6071
euclidean	significance_weighting	1.7297	1.5827	N/A	N/A	N/A	0.0000

Best ItemCF configuration by RMSE: Cosine / Weighted Average and Euclidean / Weighted Average both achieve RMSE = 0.2852. ItemCF achieves the highest catalogue Coverage (0.8214) under these two configurations, outperforming all UserCF variants. Significance weighting is consistently unstable for ItemCF — the item space is too sparse for the co-count penalty to be calibrated reliably.

10.3 Content-Based & Hybrid

Model	Similarity / Weights	Prediction	P@10	R@10	F1@10	Coverage
ContentBased	cosine (profile vs item vector)	N/A	0.1000	1.0000	0.1818	0.6071

Hybrid (40% UCF + 40% ICF + 20% CB)	Pearson (UCF) · Cosine (ICF) · Cosine (CB)	mean_centered (UCF & ICF)	0.0800	0.8000	0.1455	0.7500
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Content-Based achieves the highest Precision@10 (0.10) and perfect Recall@10 (1.0) among all models evaluated on the 30-user sample. This is a consequence of the very small item catalogue (31 items): a content-based profile that matches even one style attribute can correctly identify all held-out items. The Hybrid model trades a small drop in precision (0.08) for broader catalogue coverage (0.75 vs 0.61 for Content-Based alone).

10.4 Configuration Selected for Production (Best Models)

Based on the evaluation grid, the Hybrid recommender assembled for the web application uses the following sub-model configurations:

Sub-Model	Similarity	Prediction	Weight
UserBasedCF	Pearson	Mean-Centred	0.40
ItemBasedCF	Cosine	Mean-Centred	0.40
ContentBasedFilter	Cosine (profile)	N/A	0.20

Pearson similarity is preferred for UserCF because it corrects for individual rating-scale bias; Cosine similarity is used for ItemCF because item vectors (rating profiles across users) are generally less biased by individual rater tendencies. Mean-centred prediction consistently outperforms plain weighted average by removing per-user and per-item rating offsets.

11 Conclusion & Next Steps

11.1 What Was Achieved

- A fully working, zero-ML-framework recommendation engine implementing three fundamental paradigms (UserCF, ItemCF, ContentBased) plus a weighted hybrid, all from mathematical first principles in pure Python.
- A systematic evaluation harness that cross-validates $3 \times 3 = 9$ configurations per collaborative model (18 total CF variants) plus Content and Hybrid, using both rating-prediction metrics (RMSE, MAE) and ranking metrics (Precision@K, Recall@K, F1@K, Coverage).
- A first-class explainability layer: every model surfaces the specific evidence — neighbours, items, or features — that drove each recommendation, rendered in a collapsible UI panel.
- A polished, zero-dependency SPA with algorithm-control dropdowns, animated recommendation cards, live user search, rating history, and colour-coded model badges.
- A clean REST API with input validation, per-request model instantiation with configurable hyper-parameters, and rich item metadata augmentation.

11.2 Limitations

- The dataset contains only 31 unique items and 406 users — too small for meaningful latent-factor or deep learning approaches, but ideal for a rigorous comparison of memory-based methods.
- Models are rebuilt from scratch on every API call. For larger datasets this would require caching or a background model-serving layer.
- Significance weighting approximates co-rated item counts using a global dict (`_SW_cocount`) populated as a side-effect of similarity computation; a more principled design would pre-compute all counts up front.
- No item title or image metadata is available — recommendations display only ASINs and style attributes.

11.3 Future Work

- Matrix Factorisation / SVD: decompose the user-item matrix into latent factors using NumPy or scipy to capture hidden feature interactions.
- Implicit feedback: extend to clickstream or purchase data where no explicit rating exists, using approaches like Alternating Least Squares.

- Model caching: pre-compute and cache similarity matrices at startup (or incrementally update them) to reduce per-request latency.
- A/B testing hooks: instrument the API to log which recommendations were clicked or purchased, closing the feedback loop.
- Richer metadata: incorporate item title, brand, category hierarchy, and image embeddings to improve content-based recall.
- Authentication and user session management: allow new users to submit ratings through the UI, growing the rating matrix dynamically.

— *End of Report* —